**Topic**

This project aims to develop an interactive public health informatics dashboard that visually presents the association between oral health outcome and demographic or socioeconomic features using data from the National Health and Nutrition Examination Survey (NHANES). This project will provide a possibility to explore disparities in oral health outcomes across demographic and socioeconomic groups in the United States.

**Public health problem**
Oral diseases are one of the most common categories of chronic conditions in the United States. They strongly affect people's daily life quality, more importantly, for people with lower income, limited access to dental care, or from certain racial and ethnic minority groups. Poor oral health can lead to pain, impaired nutrition, reduced quality of life, and even affect overall health. Although national data are available through NHANES, these differences are usually shown in static reports that are not easy for policy makers to use.

**Public health informatics problem and approach**
NHANES is a great data source providing detailed data on demographics, behaviors, and clinical oral health outcome measurements. But the data lacks visualization for non-technical users to understand, which limits the data's real-life impact. Therefore, this project aims to turn these complex examination or survey data into a clear and interactive dashboard. This dashboard will allow users to explore the difference in oral health across different population groups. By using this visualization, users can associate oral health outcomes with different demographic and socioeconomic features. This can be helpful for people to investigate health disparity in the context of dental health, and further support informed decision-making for policy makers.

**Disease or Condition**

The major conditions of interest are oral health outcomes, including current untreated tooth decay, tooth loss, and periodontal disease, as measured through NHANES oral health examination and questionnaires.

**Scope**

- **Jurisdiction:** United States
- **Population:** U.S. population sampled in NHANES
- **Data source:** NHANES 2009–2010 cycle (https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2009)
- **Data linkage:** Demographic data linked to oral health examination
- **Software platform:** Python (or R if needed) for data processing and analysis; Streamlit or Dash for dashboard development

**Stakeholders**

Potential stakeholders include:

- Public health practitioners and epidemiologists
- Oral health researchers and clinicians
- Health policy analysts
- Community health organizations
- Researchers in public health and informatics
- General public

**Context**

This project focuses on the context in the United States, aiming to support efforts to improve health equity.

**Data Sources and Types**

This project will primarily use **existing public health data**, including:

- **Oral health outcome data:**
    - NHANES Oral Health Examination data, specifically dentition examination results (https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2009/DataFiles/OHXDEN_F.htm). This provides information about current untreated tooth decay, tooth loss, tooth counts, which are overall helpful to investigate oral health disparities.
- **Demographic and socioeconomic data** (https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2009/DataFiles/DEMO_F.htm)**:**
    - Age, sex, race/ethnicity, education level, income-to-poverty ratio, and household characteristics. These layers will be associated with oral health outcomes.
    - Examination, interview weights, Masked variance strata and PSU (Primary Sampling Units) variables. Because NHASES inevitably implements clustered sampling. So additional sampling information will be provided for user's validation. Such information will also be analysed to identify potential bias.

While this project does not generate new primary data, it tries to provide a more trustworthy interactive dashboard. Therefore, additional data science analysis will be applied if strong sampling bias has been identified from NHASE data.

**Tools, Resources, and Data Needs**

To complete the full project, the following resources are required:

- Publicly available NHANES data files downloaded from the CDC NHANES website
- Computational and statistical software (Python or R if needed) capable of handling complex survey designs
- Visualization and dashboard tools (Streamlit or Dash)